Academic Typesetting Portfolio

Thesis-layout demonstration

Temperature-Driven Power Demand in Distributed Sensor Networks

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LaTeX document design and typesetting sample

Portfolio demonstration only. The dataset is synthetic; this document is not an academic submission and is not affiliated with a university.

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Abstract

This document demonstrates a modular thesis-production workflow in LaTeX. It uses a reproducible, entirely synthetic dataset containing 2880 hourly observations across three example monitoring sites to illustrate data preparation, a hinge-regression model, mathematical notation, tables, and publication-style figures. The source is organized into a reusable preamble, independent chapters, appendices, bibliography, and index. All data and graphics are generated by the accompanying Python scripts. This is a typesetting sample, not a claim of empirical research or an academic qualification.

Contents

Abstract iii

List of Figures vii

List of Tables ix

1 Introduction 1
- 1.1 Motivation . . . 1
- 1.2 Research Questions . . . 1
- 1.3 Structure of the Thesis . . . 1

2 Literature and Background 3
- 2.1 Scientific Python Stack . . . 3
- 2.2 Sequence Models . . . 3
- 2.3 Estimation Theory . . . 3
- 2.4 Control Background . . . 3

3 Methodology 5
- 3.1 Data . . . 5
- 3.2 Demand Model . . . 5
- 3.3 Control Loop . . . 5
- 3.4 Pipeline . . . 5

4 Results 7
- 4.1 Exploratory Analysis . . . 7
- 4.2 Discussion . . . 7

5 Conclusion 9

A Reproducibility 11

B Derivation of the Variance Formula 13

Bibliography 15

Index **16**

List of Figures

2.1 Transformer encoder stack with residual connections. 4
2.2 Supervisory finite-state machine. 4

3.1 Closed-loop PID control. 6
3.2 Second-order step responses for three damping ratios. 6

4.1 Exploratory analysis of the sensor dataset. 8

List of Tables

4.1 Per-site summary statistics. 8

Chapter 1

Introduction

1.1 Motivation

Energy monitoring systems produce long hourly series whose behaviour is dominated by weather. Understanding the temperature–demand link supports forecasting and control [5].

1.2 Research Questions

RQ1. How strongly is power demand coupled to temperature?

RQ2. Do monitoring sites differ systematically?

RQ3. Can a feedback controller regulate the resulting demand?

1.3 Structure of the Thesis

Chapter 2 reviews tools and theory, Chapter 3 states the model and pipeline, Chapter 4 reports findings, and Chapter 5 concludes. Reproducibility material is in Chapter A.

Chapter 2

Literature and Background

2.1 Scientific Python Stack

Tabular data handling follows `pandas` [4]; plots use Matplotlib [2] and seaborn [7]. Palettes are chosen for colour-vision deficiency [8].

2.2 Sequence Models

Attention-based architectures [6] replaced recurrence in many forecasting pipelines. Figure 2.1 sketches the encoder stack, drawn natively in TikZ.

2.3 Estimation Theory

For a linear model $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ the least-squares estimator attains the Cramér–Rao bound under Gaussian noise [3].

Definition 2.1 (Hinge feature)**.** *For threshold τ the hinge feature of temperature T is* $h(T) = \max(T - \tau, 0)$.

2.4 Control Background

Classical feedback design is covered by Ogata [5]. A supervisory finite-state machine [1] can switch controller modes (Figure 2.2).

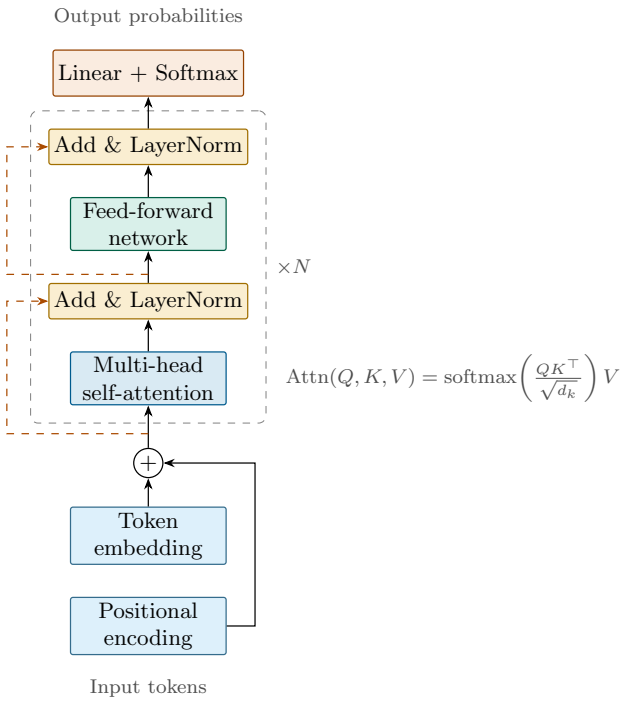


Figure 2.1: Transformer encoder stack with residual connections.

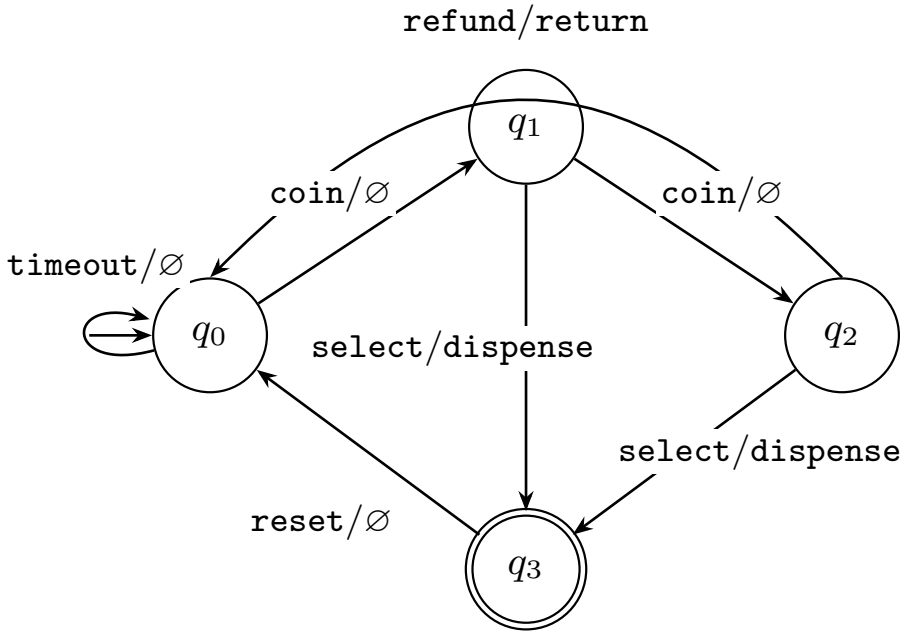


Figure 2.2: Supervisory finite-state machine.

Chapter 3

Methodology

3.1 Data

The dataset `data/sensor_experiment.csv` contains 2880 hourly rows from three sites (*synthetic*, generated with a fixed seed). Missing values (50 in total) are filled by time-ordered linear interpolation.

3.2 Demand Model

Proposition 3.1. *For $\boldsymbol{X}$ of full column rank, the estimator*

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y} \tag{3.1}$$

is unbiased with covariance $\operatorname{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\boldsymbol{X}^\top \boldsymbol{X})^{-1}$.

3.3 Control Loop

Figure 3.1 shows the closed loop; the PID law is

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)\, d\tau + K_d \dot{e}(t). \tag{3.2}$$

3.4 Pipeline

All figures come from `scripts/generate_figures.py`; see Chapter A.

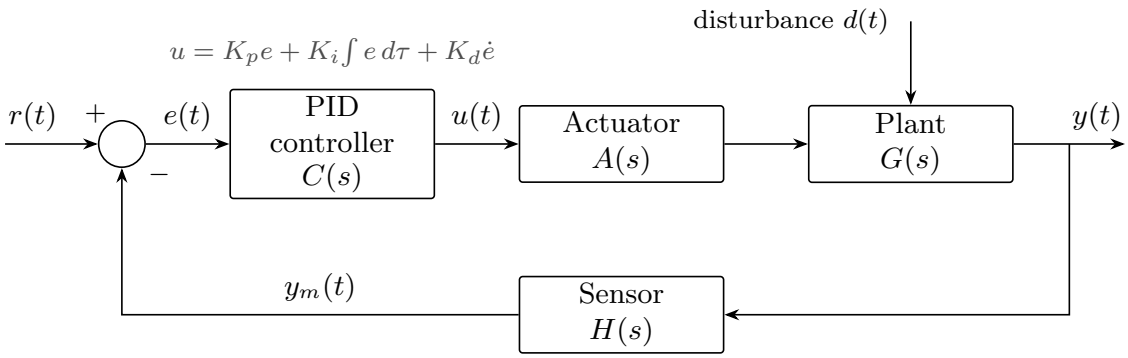


Figure 3.1: Closed-loop PID control.

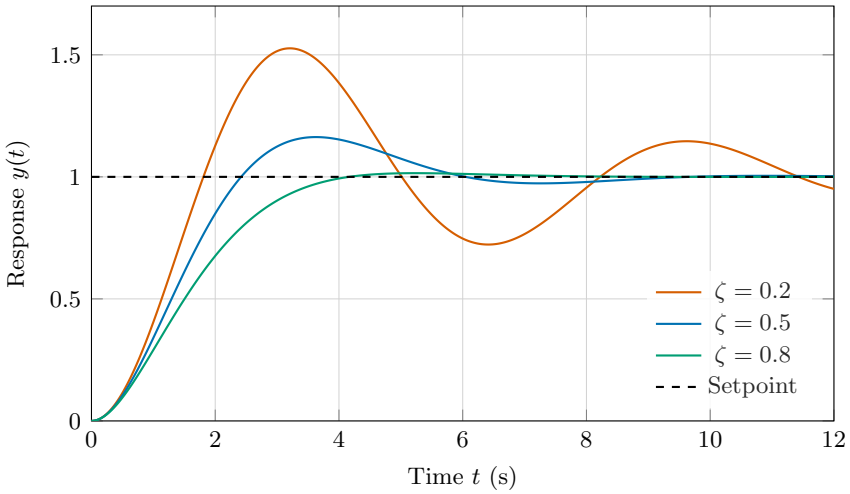


Figure 3.2: Second-order step responses for three damping ratios.

Chapter 4

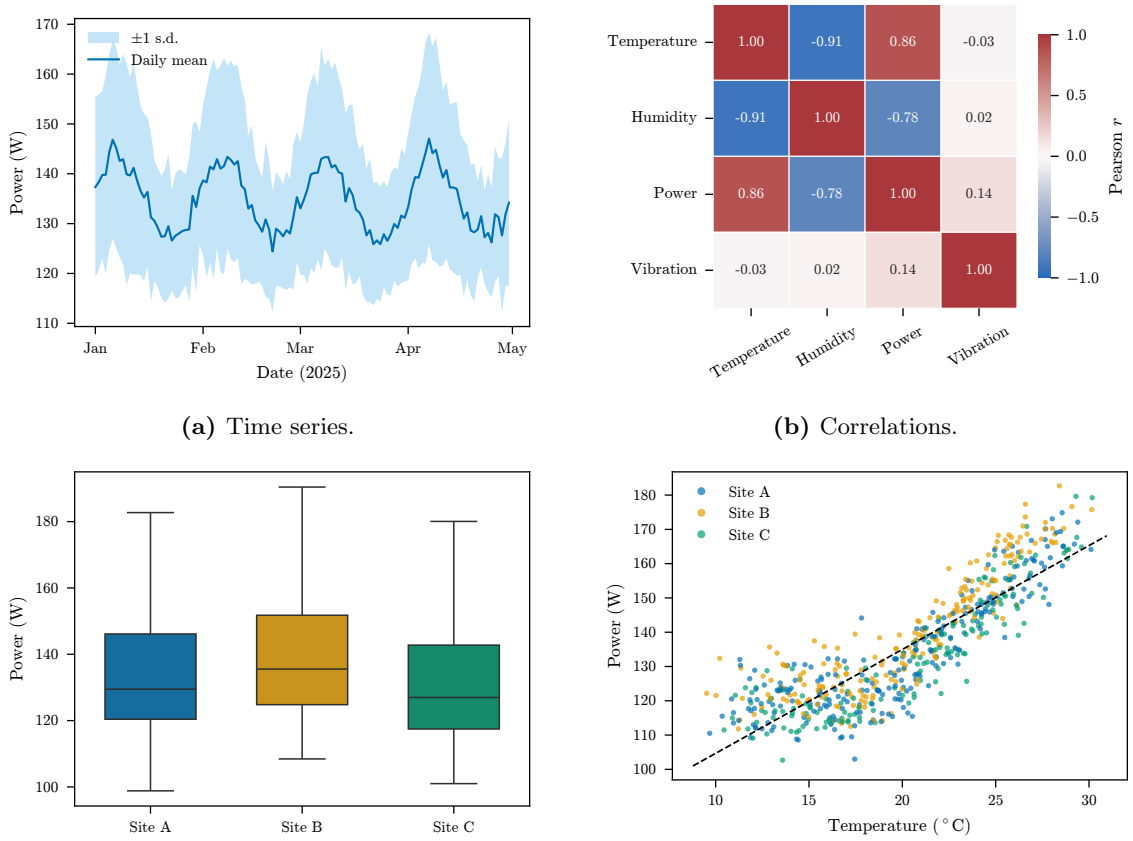
Results

4.1 Exploratory Analysis

Temperature and power correlate strongly ($r = 0.86$, Figure 4.1(b)). Site differences appear in Figure 4.1(c) and Table 4.1.

4.2 Discussion

Site B has the highest mean power, matching its positive offset in the generating process. The strong correlation justifies the hinge model of Proposition 3.1.



(a) Time series.

(b) Correlations.

(c) Per-site distribution.

(d) Power vs. temperature.

Figure 4.1: Exploratory analysis of the sensor dataset.

Table 4.1: Per-site summary statistics.

Site	n	$\bar{T}$ (°C)	$\bar{P}$ (W)	s_P (W)	$\bar{a}$ (g)
Site A	1151	19.87	133.74	16.60	0.377
Site B	989	20.18	139.18	16.92	0.385
Site C	740	19.87	131.00	16.21	0.383

Chapter 5

Conclusion

We answered RQ1–RQ3 on synthetic data: temperature explains most of the variance in demand, sites differ by a few watts, and a suitably damped PID loop regulates the response. The principal contribution is the *architecture*: modular chapters, one preamble, scripted figures.

Appendix A

Reproducibility

Regenerate data and figures, then build:

```
python scripts/make_data.py
python scripts/generate_figures.py
cd src/thesis && latexmk -pdf main.tex
```

Appendix B

Derivation of the Variance Formula

Substituting $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ into (3.1) gives $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top \boldsymbol{\varepsilon}$. Hence

$$\mathbb{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}, \tag{B.1}$$

$$\mathrm{Cov}(\hat{\boldsymbol{\beta}}) = (\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top(\sigma^2 \boldsymbol{I})\boldsymbol{X}(\boldsymbol{X}^\top \boldsymbol{X})^{-1} = \sigma^2(\boldsymbol{X}^\top \boldsymbol{X})^{-1}. \tag{B.2}$$

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Index

demand model, 1

energy monitoring, 1

least squares, 3

PID control, 3

reproducibility, 9